ARPITA PRIYADARSHANI SAMAL

Metallurgical and Materials Engineer | Casting & Solidification | ICME | Phase Field Modelling | Thermo-Calc (CALPHAD) | Light Metal Alloys

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I have always been keen for practical experiences where I can bridge computational materials science with real-world manufacturing technologies. In my industrial training journey at Center of Excellence for Light Weighting, Sundaram Clayton Limited (TVS), Chennai, SAIL, Rourkela and Aditya Aluminium & Power Plant, Lapanga I could explore and gain experience on practical challenges at industry.

As a PhD researcher at IIT MADRAS (Dept. of Metallurgical & Materials Engineering, CGPA 9.5), and as a Research and Development Intern at Center of Excellence for Light Weighting, Sundaram Clayton Limited (TVS), Chennai, I apply the ICME (Integrated Computational Materials Engineering) framework to model microstructure evolution during casting and solidification of aluminium alloys, using Phase Field Modelling and Thermo-Calc (CALPHAD) to predict phase diagrams, segregation behaviour, and grain morphology. This supports the development of lightweight alloy in automotive and aerospace applications.

Education

Degree	Institute	CGPA/%	Years
PhD – Metallurgical and Materials Engineering	Indian Institute of Technology Madras	9.5	2022-ongoing
M.Tech – Metallurgical and Materials Engineering	Indian Institute of Technology Jodhpur	9.15	2020 - 2022
B. Tech – Metallurgical and Materials Engineering	Veer Surendra Sai University of Technology, Sambalpur, Odisha	9.18	2016 - 2020
12th	Deepika E. M. School, Rourkela, Odisha	86.6	2014 - 2016
10th	Deepika E.M. School, Rourkela, Odisha	9.8	2014

Projects

PhD Project: Phase Field Modelling and microstructure development via ICME approach for casting and solidification of aluminium alloys

An ICME framework to model microstructure evolution during casting and solidification of aluminium alloys, using Phase Field Modelling and Thermo-Calc (CALPHAD) to predict phase diagrams, segregation behaviour, and grain morphology.

Supervisors: Prof. Ranjit Bauri and Prof. Gandham Phanikumar

M.Tech Project: Role of microstructure and bio-corrosion behavior of Mg rich Mg-Zn-Nd alloys

The work incorporated neodymium additions (upto 5%) in Mg-4Zn alloy in order to improve mechanical and corrosionresistance property which found its application in biomedical implants.

Supervisor: Prof. Ravi K.R.

B.Tech Project: Effect of Sintering Temperature and Reinforcement Composition on Microstructure and Mechanical properties of Al-SiC-Fly Ash Hybrid Composite using Powder Metallurgy Route

Fabricated Al-SiC-Fly Ash hybrid composites via powder metallurgy, varying sintering temperature to optimize hardness, density, and compressive strength – work published in Springer (Processing and Characterization of Materials, 2023, DOI: 10.1007/978-981-99-5509-1_8).

Supervisor: Dr. Dinesh Kumar Mishra

Industrial Training and Internships

Research Intern at Lightweight Alloy Development, Center of Excellence for Lightweighting

Sep 2023 - May 2026

Sundaram-Clayton Limited (TVS)
Chennai

ICME-driven microstructure modelling for lightweight aluminium alloys in automotive applications. Phase Field Modelling and Thermo-Calc (CALPHAD) have been used to simulate solidification microstructure evolution with experimental validation (SEM, TEM, HRTEM, XRD), supporting alloy design for casting applications.

Summer Research Training Programme

Jun 2020 - Aug 2020

CSIR-Institute of Minerals and Materials Technology
Bhubaneswar

Prepared iron aluminide powder for thermal spray coating applications under Principal Scientist Dr. Mayadhar Debata, gained expertise in powder synthesis and characterisation techniques.

Vocational training in Cast House, Smelter Plant

Jun 2019 - Jul 2019

Aditya Aluminium and Power Plant
Lapanga, Sambalpur

Observed and documented end-to-end aluminium casting operations at a commercial smelter, delivered a technical presentation on cast house process flow that was formally recognized by Aditya Aluminium management.

Research intern

Dec 2018 - Dec 2018

National Institute of Technology, Rourkela

Collaborative research project on process metallurgy under the guidance of Dr. Snehanshu Pal.

Vocational training in Blast Furnace

May 2018 - Jun 2018

SAIL, Rourkela Steel Plant
Rourkela

Studied ironmaking and steelmaking in one of India's largest integrated steel plants, produced a blast furnace technical report recognized by RSP management.

Accomplishments

- Poster presentation in the 20th International Conference on Aluminium Alloys (ICAA20-2026), Berlin, Germany, 13th - 17th September, 2026.
- Poster presentation in 7th International Conference on Processing and Characterization of Materials (ICPCM-2025), NIT Rourkela, 12nd - 13th December, 2025.
- Poster presentation in the 30th International Conference on Processing of Advanced Materials and Fabrication of Products (PAMFP-2024), IIT Hyderabad, 11st - 13th December, 2024.
- Poster presentation in the 28th International Conference on Non-ferrous Metals - 2024 (ICNFM - 2024),

Bhubaneswar, 12nd - 13th July, 2024.

- Poster presentation in the National Symposium - "Recent Trends in Material Science", AMALGAM, IIT MADRAS, 3rd March, 2024.
- Paper presentation in the "2nd Virtual National Conference on Advancement in Materials Processing Technology" at NIT, Jamshedpur, 31st October - 2nd November, 2020.
- Paper presentation in "Webinar for Young Researchers in Metallurgical & Materials Science", CSIR - National Metallurgical Laboratory, Jamshedpur, 12th - 14th August, 2020.
- Paper presentation in "International Conference on Innovations in Aluminium Technology" at IIM, Angul, 8th - 9th February,2020.
- Paper presentation in the "4th International Conference on Advances in Steel, Power and Construction Technology" at OP Jindal University, Raigarh during 21st - 22nd November, 2019.
- Poster presentation entitled in TEQIP-III sponsored "NCFAM - National Conference on Frontiers in Advanced Materials" at VSSUT, Burla, Odisha during 27th - 28th July, 2019.

Awards

- **Silver medal** for Best Academic Performance, M.Tech Materials Engineering Program
- **1st position** in Poster presentation, International Conference on Non-Ferrous Metals (ICNFM-2024), Bhubaneswar
- **2nd position** in Poster presentation, PAMFP-2024 (30th International Conference on Advanced Materials & Fabrication), IIT Hyderabad
- **Winner** of OLA Sankalp 2024 Thought Paper Contest
- **Winner** of Digital Clash Competition, AIRSS, IIT Madras (2024)
- **2nd prize** in Poster Making Competition, IIT MADRAS (2024)
- **1st position** in Sri Vasudevan Quiz (Meta Quiz), Amalgam, IIT Madras (2023)
- **1st position** in Online Technical Presentation, Diamond Jubilee Year Celebration, NIT Rourkela (2021)
- **1st position** in Online Quiz Contest, Diamond Jubilee Year Celebration, NIT Rourkela (2021)
- **2nd position** in Poster presentation, IIM Reminiscence, organised by IIEST Shibpur (2020)

Certifications

- Hands-on Workshop on “Computational Thermodynamics” — IIT KANPUR (Apr - 2025).
- Hands-on Workshop on “Rietveld Refinement Analysis” – IIT MADRAS (Jun 2024).
- Workshop on “Advanced Materials Characterization” – MNIT JAIPUR (Mar 2024).
- Hands-on Workshop on “MicroSim” – IIT MADRAS (Jan 2024).
- SPARC International Workshop on “Progress in Structural Aluminium Die Casting: Alloy, Process and Joining”, (Online - Sep 2023).
- Thermo-Calc Software Training — Bhanu Scientific Systems Pvt Ltd (BSSPL), Hyderabad (May 2023).
- NPTEL certifcation course on "Aluminium based Alloys and Metal Matrix Composites" (July-Oct 2022).
- Workshop on “Powder Metallurgy: Fundamentals, Applications and Advancement” – NIT ROURKELA (Jul 2019).
- TEQIP-III sponsored National Workshop "SPCAM - Synthesis, Processing and Characterization of Advanced Materials" - VSSUT, Sambalpur (Apr 2019).
- TEQIP-III sponsored National Workshop "LSDMB - Large Scale Dependent Materials Behaviour" - VSSUT, Sambalpur (Feb 2019).
- Training on "C Programming and C++", Manipal Institute of Computer Education, Odisha (Dec 2017-Mar 2018).
- TEQIP-III sponsored National Workshop "EXPERIMENTS IN NON-DESTRUCTIVE TESTING" - VSSUT, Sambalpur (Feb 2018).

Teaching Assistantship

- "Stability of Microstructure" course.
- NPTEL Online Certification course on "Theory and Practice of Non Destructive Testing".
- "Structure of Materials" course.

Skills

- **Computational**: Thermo-Calc (CALPHAD) · Phase Field Modelling · ICME Framework · Solidification Modelling · Microstructure Prediction
- **Characterization**: SEM · TEM · HRTEM · XRD · Optical Microscopy · ImageJ · Metallography (ferrous & non-ferrous)
- **Mechanical Testing**: Tensile Testing · Vickers / Brinell Hardness · Compressive Strength · Density Measurement
- **Software & Programming:** · Linux Operating System · Windows Operating System · LateX · Origin Pro · C Programming · C++ · Microsoft Office · Jupyter Notebook - SageMath
- **Process Knowledge:** Casting & Solidification · Powder Metallurgy · Aluminium Smelting · Blast Furnace Ironmaking · Alloy Design · Lightweight Alloys

Language

- English
- Hindi
- Odia

Declaration

I hereby declare that the information stated above is true to the best of my knowledge.

Dt – 13/10/2026

Arpita Priyadarshani Samal